Networking and IoT Solutions (Vernetzung und IoT-Lösungen) — 10-Day Edition

Day 1 · Introduction, Pre-Study Rehash & The Master Class

Make things talk to each other — then prove the system still works when something dies.

whoami

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Who are U?

*(Write this down — it is the **start of your in-class portfolio notes**, in addition to the pre-study.)*

- Who has **programming** experience? In which languages?
- Who has used a **microcontroller** — Arduino, ESP32/ESP8266, Raspberry Pi, …?
- Who has **configured a router**, set an IP address by hand, run `ping` / `traceroute`?
- Who has used **Wireshark**, `tcpdump`, or a packet capture?
- Who has used **MQTT**? **Node-RED**? **Python**? The **command line**? **Git/GitHub**?
- **What would you like to connect to what?**
- **What do you expect from this class?**

This is an experiment!

- Teaching with **modern education principles** — challenge- and project-based
- You are supposed to **learn and explore**
- Put **you** front and center; give the content **meaning**
- **You** bring a meaningful challenge
- This is a **first-time course**: parts of it are still being written, with you

How the class works — the deal we make

- Every module **begins with its learning goals** — that is the *contract*: demonstrate the goals and you have met the module
- The **tasks are a draft** — you may modify, replace, or extend them, as long as you reach the goals
- Tasks are marked with tiers; the **syllabus** explains what ★ **Core** and ◇ **Stretcher** mean for you
- A moving ***DRAFT BOUNDARY*** marks what we are still designing together — your input shapes it
- **AI drafts, you verify** — against a capture, a datasheet, a routing table, or a running system
- **Already know a topic?** Skip the task, document why, and show the skill in a quick check

(Let's open the syllabus and Module 1 together now.)

Class logistics (short version)

- **No separate exercise sheets** — but expect at least ~6 hours/week of self-study to finish labs, portfolio, and project work
- **10 working days / 48 contact hours, one module per day**; session lengths vary (typically two sessions of 2 × 45 min, longer blocks on some days)
- You work in small **Task Pods of two or three** (three by default in a large cohort). For the **final project, two pods merge into a Project Team (4–6)**
- In-class exercises are initially **individual**
- **Ongoing feedback** in the lab — talk to us!
- **Assessment, checkpoints, task tiers, and the grade scale live in the syllabus** — your **portfolio** is the record

Everything lives in the online workbook

Central entry point: **the course repository**.

- **Module index & roadmap** — every module and the day-by-day plan
- **Syllabus** — schedule, objectives, assessment, tools, policies
- **Pre-study guide** — Module 0
- **Module 1** — today's module
- **Resource bank** — cheat sheets, RFCs, troubleshooting, story canon
- **Master Class workbook** — the hands-on steps for today

(Let's open the workbook on screen now — this is the material we will actually use.)

Today's path

1. **Who we are** — and how this class works
2. **Form Task Pods** and check in on the pre-study
3. **The story** — a dashboard that needed a cloud
4. **Pod study** — the IoT computing continuum and what layers cost
5. **The Master Class** — a real sensor, broker, and dashboard, end to end
6. **Measure, then pitch** — the overhead, and the system you just built

Form your Task Pod — who do you build with?

1. On a note, write **your skills** (networking, Linux, Python, hardware, documentation, design, …) and **your expectations / what you want to build**.
2. No pod yet? Find **one or two other unpaired people** (three-person pods are the default in a large cohort; pairs are welcome). Compare notes and check whether you **complement** each other — not just “same as me”.
3. Exchange contacts and a first idea today. If you already have a pod, share your notes so the room knows who can help with what.
4. Agree on how you keep in touch and how you split roles.

Why “Task Pod”? In orchestration a *pod* is the smallest deployable unit — a few tightly coupled containers that are scheduled, run, and retired together. Your pod is that, for people. Two pods later merge into a **Project Team**.

Portfolio check — before we touch hardware

1. Open your portfolio repository (created from the iot-portfolio-template; instructor added as collaborator).
2. Today's entry is a **section in** `Module01/README.md`, not a new file — files are fine too, sections are simpler to read.
3. Write `## Who I am` (your *Who are U?* answers + expectations) and `## My pod` (names, skills, first idea).
4. During the day you add `## Pre-study artefact`, `## Study log`, and (optional) `## Story seeds` — shared notes are kept **once per pod**; the others link to that section and add one sentence.
5. Commit and push — a small commit with a clear message.
6. Keep `PreStudy/` where it is: this is an **additional** in-class entry, not a replacement.

Pre-study check-in

(In your pod. Two minutes per person.)

1. Show **one artefact** from Module 0 — your packet-path sketch, the `traceroute` with the private-to-public transition marked, or your first MQTT screenshot.
2. Explain **one thing you understood only after doing it**.
3. Name **one question you still have** — park it on the board.

(We pick the questions up during the day; several of them are today's lesson.)

The story — The Dashboard That Needed a Cloud

Stadtwerke Gütersloh operates district-heating substations across the city.

Petra Sommer, network operations, wants one simple thing: the **temperature and pressure of each substation**, updated every few seconds, visible in the control room and on a technician's phone.

Cirrus Cloud Systems proposes the usual stack: a cloud broker, a managed database, a monthly fee — and a firmware update for every controller.

The two questions we do not answer today

> **Petra:** *“The data is produced in the basement of a building we own, and the person who reads it is standing in the same building. Why does it need a round trip through the internet?”*

> **Malik Yilmaz, field engineer:** *“And when the uplink is down — what does my dashboard show me?”*

By the end of today you will have built the loop those two questions are about. Whether the answer is *“the cloud”* or *“a box in the basement”* is yours to defend.

What you will build today

```
[ M5StickC ]  --Wi-Fi-->  [ Broker ]  -->  [ Node-RED ]
 button, LED,               (MQTT)          gauge, chart,
 DS18B20 temp                               +/temp heatmap

... and the other way:  Node-RED  -->  Broker  -->  M5StickC LED
... and the whole room on one chart with +/temp
```

- A **sensor** that publishes, a **broker** that carries, a **flow** that reacts
- A **button** that switches hardware in another part of the room
- A **dashboard** a stranger could read without you talking

The Master Class — five phases

Follow the **Master Class workbook**; Module 1 adds the story and the evidence.

1. **First contact** — room Wi-Fi, your Node-RED, inject → debug
2. **Your stick answers** — button, LED, screen text
3. **The domino chain** — your button switches your neighbour's LED
4. **A real sensor** — wire the DS18B20, watch an over-the-air deploy
5. **The control room** — gauge, chart, and the whole room with `+/temp`

Then: measure what it cost, and pitch the system.

The order is a suggestion

- The phases are a **path**, not a timetable. Take a different route if your pod can still **prove the goals**.
- **The instructor demos live in Node-RED** — ask for the demo you need, when you need it, instead of waiting.
- Stuck on wiring? Swap to a flow task. Bored of flows? Wire the sensor. The room is a lab, not a queue.
- **Write down what you changed and why.** The checkpoint question is always *“what did you choose, and what did it cost you?”*

Where you are

What	Where to find it
Room Wi-Fi	`iotempire-b1300-5g` (laptop) · `iotempire-b1300` (2.4 GHz, sticks) · password `iotempire`
Your Node-RED	`http://<instructor-host>:18nn/` — exact host and port on the projector
Login	`admin` + your pod password from the projector
Broker	`broker.internal`, port `1883` — already configured
Your node	the M5StickC labeled with your pod's two-digit ID

Hardware per pod: 1× M5StickC, 1× USB-C cable, 1× Dallas DS18B20 + 3 jumpers.

Your roles — and rotate them

- **Driver** — laptop, Node-RED canvas, deploys
- **Navigator** — workbook open, topics and pinouts, keeps the notes
- **Documenter** — photos, screenshots, the one-line “what just happened”

Every member should have **published to a topic** and **wired a jumper** at least once today. Swap after each phase.

> **Who owns the machine?** Your Node-RED runs on the instructor's host. That is why there is nothing to install — and why the whole room shares one point of failure. Keep that for the pitch.

Story hooks — add a chapter

The story is not finished. Each phase has a chapter you can extend with **what you just saw**:

- **The lamp test** — a basement button lights a lamp in the control room.
- **“I am here”** — Malik's button, and the light that says the door is open.
- **The alarm relay chain** — each substation rings the next; the last rings the first. *What happens when pod 07 goes for coffee?*
- **Somebody leaned on the pipe** — pinch the sensor: a burst pipe. Pull the jumper: `-127`. The sensor is not lying.
- **Petra's accountant** — twenty bytes of temperature in a frame that is mostly paperwork.
- **The district is the room** — the `+/temp` chart, and the corner that is always colder.

Add **one line of your own** to your portfolio and bring it to the pitch.

What to keep today

- `## Who I am` and `## My pod` in `Module01/README.md`
- One pre-study artefact, linked with one sentence
- A screenshot of inject → debug, and your button/LED flow
- Your **domino** flow as `.json`, and what its weakest link is
- A photo of the wired stick and `<id>/temp` arriving
- The gauge, the chart, and the `+/temp` heatmap
- Your **overhead table** with the assumptions behind it
- The **continuum sketch**, the two-minute pitch, and the weakest-link sentence

Before you leave

1. **One line for the story** — what did your system do that surprised you?
2. **Commit** — small commit, clear message, link your pod mates where you did not work alone.
3. **One question for tomorrow** — leave it on the board.

Next: Module 2 — Local Networking & OpenWrt Gateways. Subnetting, DHCP, DNS, NAT, and the first capture at the choke point.

Questions?